

Megan Coyne

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Education

Bachelor of Science in Computer Science

May 2026

University of Dayton | Dayton, OH

Honors with Distinction, Dual Major – Mathematics, Computer Science, 3.88 GPA

Undergraduate Research

Thesis applying Extreme Value Theory to predict municipal financial risks driven by industrial instability.

Skills

Python | Java | Docker | HTML | Golang | GIT | Microsoft Excel | Agile Programming Methodologies | Postman |
REST APIs | SQL | AngularJs | JUnit5

Work Experience

Software Engineer | University of Dayton Research Institute, Dayton, OH

October 2022 – Present

- Develop software tools using Golang, C++, and Python to build and maintain projects for the Autonomy Capability Team, focused on operationalizing AI for the Air Force.
- Create, maintain, and test GitLab CI/CD pipelines for artificial intelligence projects.
- Write Dockerfiles and manage software requirements for containerized programs.

Teaching Assistant | University of Dayton Mathematics Department, Dayton, OH

August 2025 – Present

- Assist with instruction for Calculus III, including attending lectures, grading assignments, and hosting structured review sessions.

Enterprise Technology Intern | Ford Motor Company, Dearborn, MI

May 2025 – August 2025

- Built a Go backend for an interactive app with layered diagrams and clickable Markdown zones.
- Developed REST controllers, integrated GitHub App data, and presented the tool to Ford's Lead Architect.
- Selected as 2025 Intern Social Ambassador, representing the intern experience through curated content.

Enterprise Technology Intern | Ford Motor Company, Dearborn, MI

May 2024 – August 2024

- Created a reference app used by 1,000+ engineers to navigate Ford's internal coding conventions.
- Built and tested APIs using Java Spring Boot, Postman, and JUnit5.
- Deployed the application on Google Cloud Run and configured secure access via ADFS.

Research Associate | Air Force Research Lab, Wright Patterson AFB, OH

May 2023 – August 2023

- Created a graphical user interface for a change blindness study using Python and Scrum methodologies.
 - Analyzed experiment data to further AI research and the development of the Air Space Cognitive Engine.
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Campus Involvement

Math Club | President

August 2022 – Present

- Organize lectures on math research and new discoveries, as well as host math-themed board game nights.
- Plan the Putnam Exam, coordinate the Math Club Picnic, and manage the High School Math Competition.

Sigma Kappa | Vice President of Member Integrity

January 2023 – Present

- Lead social and service initiatives, including activities raising awareness for Alzheimer's disease.
- Liaise between members and national leadership to support conflict resolution and uphold standards.

Flyer Consulting | Consultant

January 2024 – Present

- Provide high quality, thorough business solutions to non-profits and small businesses in the Dayton area.
- Advise clients on technical implementations to enhance business operations and efficiency.

Nebraska Conference for Women in Math - UD Representative

January 2024